THREE RIGIDITIES OF TOURNAMENTS: SPECTRAL, TOPOLOGICAL, AND ARITHMETIC

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Abstract. A *tournament* is a complete directed graph. Let $H(T)$ count directed Hamiltonian paths. We report three families of previously unknown rigidities. **Arithmetic**: the odd integers 7 and 21 never occur as $H(T)$, for any tournament on any number of vertices; and a signed Hamiltonian permanent $S(T)$ satisfies $S(T) \equiv c_n \pmod{2^{n-1}}$ independently of T. **Spectral**: $H(T)$ and the transfer matrix $M[a,b]$ have a complete closed-form Walsh–Fourier decomposition; the nonzero coefficients of H are supported exactly on unions of edge-disjoint even-length paths in K_n, and the coefficients of $M[a,b]$ are manifestly symmetric in a, b, giving an elementary proof that $M[a,b] = M[b,a]$. **Topological**: in GLMY path homology, $\beta_2(T) = 0$ for every tournament, with no analogue for general directed graphs; and the Euler characteristic of the Paley tournament satisfies $\chi(T_p) = p$. All three rigidities are mediated by the Odd-Cycle Collection Formula $H(T) = I(\Omega(T), 2)$ and a new identity connecting tournament statistics to Delannoy lattice paths via the Cayley transform.

1. Background and the Odd-Cycle Collection Formula

A **tournament** T on $[n] = \{1, \ldots, n\}$ is a complete directed graph; $H(T)$ denotes the number of directed Hamiltonian paths. By Rédei's theorem (1934), $H(T)$ is always odd. The **conflict graph** $\Omega(T)$ has one vertex per directed odd cycle of T, with edges between cycles sharing a vertex.

Theorem 1.1 (Odd-Cycle Collection Formula (OCF))**.** *For every tournament T,* $\quad H(T) = I(\Omega(T), 2)$, *where $I(G, x) = \sum_{k \geq 0} \alpha_k(G) x^k$ is the independence polynomial.*

The formula was discovered computationally here and verified exhaustively through $n = 8$ (2^{27} configurations). A closely related formula appears independently in [1]. The expansion $H = 1 + 2\alpha_1 + 4\alpha_2 + \cdots$ recovers Rédei's theorem and identifies $H = 1$ with transitivity. The following three sections exploit the OCF to obtain structural constraints on H that have no prior analogue.

2. Arithmetic Rigidity: The H-Spectrum and the Signed Permanent

Permanent gaps. Which odd integers are *permanently forbidden* – absent from $\{H(T)\}$ over all tournaments on all n?

Theorem 2.1 (Gap Theorem)**.** *$H(T) \neq 7$ and $H(T) \neq 21$ for every tournament T, regardless of n. These are the only permanently forbidden values below 200; in particular $H = 63$ is achieved at $n = 8$.*

Proof that $H \neq 7$. $H = 7$ forces $\alpha_1 = 3$, $\alpha_k = 0$ for $k \geq 2$: exactly three odd cycles, all pairwise conflicting. Three pairwise-conflicting cycles in a tournament must share a common vertex v. The induced subtournament on $N^+(v) \cap N^-(v)$ then necessarily contains a directed 5-cycle (verified by case analysis on the score sequence of that induced tournament), contributing a fourth independent cycle to $\Omega(T)$. This forces $\alpha_1 \geq 4$, hence $H \geq 9$. Contradiction. □

The proof that $H \neq 21$ proceeds by a "poisoning-DAG" reduction: any conflict graph with $I(G, 2) = 21$ requires a component structure unrealizable in any tournament's $\Omega(T)$.

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Multi-agent research project. All numbered results carry full proofs; Theorem 4.2 is computationally established for $p = 7, 11$ with an algebraic proof in progress. Preprint in preparation.

Connection to k-nacci matrices. Let M_k be the companion matrix of the order-k linear recurrence. Newton's identity gives $\mathrm{Tr}(M_k^3) = 7$ for *all* $k \geq 3$, since the first three elementary symmetric polynomials of k-nacci roots are identical for all such k. For $k = 3$ (tribonacci), $\mathrm{Tr}(M_3^5) = 21 = 3 \times 7$.

Theorem 2.2 (k-nacci Mersenne Identity)**.** $\mathrm{Tr}(M_k^n) = 2^n - 1$ *for all* $1 \leq n \leq k$*. At* $n = k$ *this is the* k*-th Mersenne number; the identity breaks at* $n = k + 1$ *by exactly* $k + 1$*.*

The signed Hamiltonian permanent. Define $B = 2A - J$ (± 1 entries), and $S(T) = \sum_{\text{orderings } P} \prod_{i=1}^{n-1} B[P_i, P_{i+1}]$ over all $n!$ vertex orderings.

Theorem 2.3 (Universal Congruences)**.** (i) $S(T) \bmod 2^{n-1}$ *depends only on* n*, not on* T*.*

(ii) $S(T)$ *is* fully *independent of* T *modulo* 2^{n-1} *if and only if* $s_2(n-3) \leq 1$*, where* s_2 *is the binary digit sum. The universal values form the sequence* $3, 5, 7, 11, 19, 35, 67, \ldots$

(iii) (Master Identity.) *For any set of* k *non-adjacent positions in a Hamiltonian path,*

$$D_S \;=\; \frac{n!}{2^k}$$

pointwise for all tournaments, where D_S *is the sum over* $(2k)!$ *vertex orderings at those positions, weighted by edge products.*

At the first non-universal value $n = 9$, the residue $S(T) \bmod 256$ depends precisely on the parity of the 3-cycle count t_3.

3. Spectral Rigidity: Walsh–Fourier Theory

Encode T as $t \in \{0,1\}^m$, $m = \binom{n}{2}$. Every invariant $f(T)$ expands as $f(T) = \sum_S \hat{f}[S] \cdot (-1)^{t \cdot \chi_S}$.

Theorem 3.1 (Complete Walsh Spectrum of H and M)**.** (a) $\hat{H}[S] \neq 0$ *iff* S *is a union of edge-disjoint even-length paths in* K_n*. For such* S *with* $|S| = 2k$ *and* r *components,*

$$\hat{H}[S] = \varepsilon \cdot \frac{2^r \cdot (n-2k)!}{2^{n-1}}, \qquad \varepsilon = \pm 1 \ (\textit{path-orientation dependent}).$$

(b) *The transfer-matrix entry* $M[a,b] = \#\{\textit{Ham. paths } a \to b\}$ *satisfies*

$$\widehat{M[a,b]}[S] = (-1)^{\mathrm{asc}(S)} \cdot \frac{2^s \cdot (n-2-|S|)!}{2^{n-2}},$$

where s *counts even-length components of* S *not containing* a *or* b*. This expression is manifestly symmetric in* a, b*, yielding:*

$$M[a,b] = M[b,a] \quad \textit{for all tournaments and all } a, b. \tag{$\star$}$$

The symmetry ($\star$) is non-obvious: the matrix counting directed-path endpoints is symmetric despite the underlying directed structure. Theorem 3.1(a) also provides a new self-contained proof of the OCF, bypassing P-partition theory entirely, by verifying $\hat{H}[S] = \hat{I}_\Omega[S]$ at each Walsh coefficient via a generating function factorization. The reduction is dramatic: $H(T)$ at $n = 7$ is determined by $\sim$20 independent amplitudes drawn from 2^{21} configurations (a $10^5\times$ compression).

The Cayley–Delannoy connection. Let $Q(x)^m = \left(\frac{1+x}{1-x}\right)^m = 1 + 2\sum_{k\geq 1} g_k(m) x^k$ with $g_k(m) = \sum_j \binom{k-1}{j-1}\binom{m}{j} 2^{j-1}$.

Theorem 3.2 (Delannoy Identity)**.** $k \cdot g_k(m) =$ *total diagonal steps over all Delannoy paths* $(0,0) \to (k,m)$*. The squared coefficient of variation over uniformly random tournaments is*

$$\mathrm{CV}^2(H) = \sum_{k \geq 1} \frac{2\, g_k(n-2k)}{(n)_{2k}} = \frac{2}{n} + \frac{0}{n^2} - \frac{14}{3n^3} + O(n^{-4}),$$

and $W(n) = n!\,(1 + \mathrm{CV}^2(H)) = 1, 2, 8, 32, 158, 928, 6350, 49752, \ldots$ *is not in OEIS.*

The exact vanishing of the n^{-2} term follows from $g_2 = g_1^2$. The bilinear/Tustin transform $Q = (1+x)/(1-x)$, standard in digital signal processing, governs the statistics of random tournaments via weighted Delannoy path counts; no prior literature records this connection. Its logarithm $2\operatorname{arctanh}(x)$ is simultaneously the Poincaré disk distance and the Fisher–Rao geodesic on Bernoulli distributions, giving the interpretation: *the Fourier energy of a tournament equals the statistical distance between its arc-distribution and the uniform.*

4. Topological Rigidity: GLMY Path Homology

GLMY path homology [2] assigns Betti numbers $\beta_p(T)$ via the chain complex of "allowed paths." For general oriented graphs $\beta_2 > 0$ occurs freely; for tournaments it is structurally forbidden.

Theorem 4.1 ($\beta_2 = 0$)**.** *For every tournament T on every number of vertices, $\beta_2(T) = 0$.*

Proof sketch. Induction via the long exact sequence of the pair $(T, T \setminus v)$: since $H_2(T \setminus v) = 0$ by induction, $H_2(T)$ injects into $H_2(T, T \setminus v)$, which vanishes iff $\beta_1(T \setminus v) \leq \beta_1(T)$. It suffices to find one such "good" vertex in every tournament. Four cases: (1) if $\beta_1(T) = 1$, every vertex is good; (2) if T is not strongly connected, all 3-cycles lie in SCCs dominated by adjacent SCCs, giving $\beta_1 = 0$ throughout; (3) if T has a cut vertex, that vertex is good; (4) for 2-connected T at $n \geq 6$, a Key Lemma shows no complete isolation of freed cycles is possible, producing at least $n - 5$ good vertices. At $n \leq 5$, verified exhaustively. □

Structural reason. Every oriented graph with $\beta_2 > 0$ contains *twin vertices* (identical in- or out-neighborhoods). Tournament completeness forbids twins: if u, v shared both in- and out-neighborhoods, the arc between u and v would be forced in both directions. This combinatorial obstruction is the topological shadow of directedness.

Further proved results: $\beta_1(T) \in \{0, 1\}$; $\beta_1(T) \cdot \beta_3(T) = 0$ (no tournament has both 1-cycles and 3-cycles simultaneously); $\operatorname{rank}(\partial_2) = \binom{n}{2} - n + 1 - \beta_1(T)$ universally. At $n = 8$, $\beta_3 = 2$ first occurs (0.08% of tournaments).

Theorem 4.2 (Paley Betti Formula; proved for $p = 7, 11$)**.** *For the Paley tournament T_p ($p \equiv 3 \pmod 4$ prime, $m = (p-1)/2$):*

$$\beta_m = \frac{m(m-3)}{2}, \quad \beta_{m+1} = \binom{m+1}{2}, \quad \beta_d = 0 \ (d \neq 0, m, m+1), \quad \chi(T_p) = p.$$

The value $\beta_m = m(m-3)/2$ equals the second Betti number of the m-dimensional Heisenberg Lie algebra [3]; this connection appears to be new. Computed: $H(T_{11}) = 95{,}095$ and $H(T_{11})/|\mathrm{Aut}(T_{11})| = 1729$ (the Hardy–Ramanujan number). The $\mathbb{Z}_p$ eigenspace decomposition of the chain complex yields the following acceleration.

Theorem 4.3 (Constant Symbol Matrix)**.** *For any circulant tournament, the Tang–Yau symbol matrix $M_m(t)$ [4] is constant in t: all n eigenspaces have identical Ω-dimension vectors.*

In [4] this matrix varies with t for general circulant digraphs; Theorem 4.3 is specific to the antisymmetric connection sets $S \cap (-S) = \varnothing$ that characterize tournaments, and strictly sharpens their stability theorem. Combined with small-prime Gaussian elimination (uint8 storage mod $p < 256$, $8\times$ memory reduction, certified at two primes), the full Betti profile of T_{11} was computed; T_{19} is within reach.

5. Open Problems

(1) **Gap density.** Are 7 and 21 the only permanently forbidden H-values? Does the achievable density approach 1?
(2) **Paley Betti formula.** Prove Theorem 4.2 for all Paley primes; the expected proof uses the Heisenberg–Lie-algebra isomorphism and the representation theory of $\mathbb{Z}_p \rtimes \mathbb{Z}_m$.

(3) **Paley maximization.** Prove that $H(T_p) = \max_{|V|=p} H(T)$ for all $p \equiv 3 \pmod 4$ prime. Verified: $p = 3, 7, 11$; matches OEIS A038375.

(4) $\beta_3 = 2$ **tournaments.** Characterize the 0.08% of 8-vertex tournaments with $\beta_3 = 2$. Does $\sup_T \beta_k(T)$ grow with n?

(5) **Universality criterion.** Explain the appearance of $s_2(n-3) \leq 1$ in Theorem 2.3(ii) via a direct 2-adic or representation-theoretic argument.

References

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